

Business Case: Wireshark in Cyber Security

Network Traffic Analysis and Protocol Investigation using Wireshark

1.Introduction

Wireshark is a widely used network protocol analyzer that enables real-time packet capture and deep inspection of wireshark traffic, it is commonly used by network administrators , SOC analysts, and cybersecurity professionals to monitor communication between systems, troubleshoot connectivity issues,and analyze suspicious network behavior.

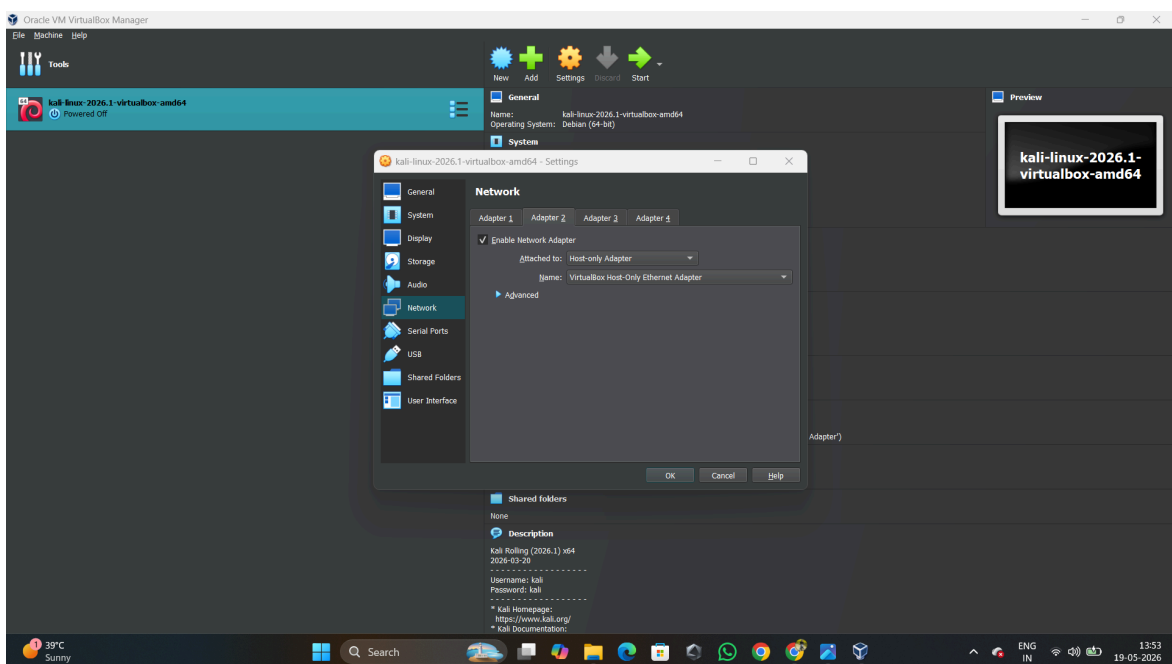
In this project , Wireshark was used to analyze communication between a Windows host system and a Kali Linux virtual machine in a controlled lab environment. Multiple protocols including ICMP, TCP, DNS and HTTP were captured and inspected to understand packet level communication and traffic behavior.

2.Problem Statement

Organizations often face challenges in monitoring and securing internal network traffic. Limited visibility into communication between systems can make it difficult to identify connectivity issues, suspicious activity, and insecure communication protocols.

The major challenges addressed in this project include:

- Monitoring communication between systems in real time
- Understanding protocol-level network behavior
- Identifying insecure or unencrypted traffic
- Troubleshooting communication and connectivity issues
- Improving visibility into internal network traffic



1.VirtualBox network configuration

2.Host-Only adapter setup

3.Solution Overview

Wireshark was deployed in the virtual lab environment to capture and analyze network packets exchanged between Kali Linux and Windows systems.

The project focused on:

- Capturing live traffic between systems
- Analyzing ICMP, TCP, DNS, and HTTP protocols
- Monitoring communication behavior between hosts
- Inspecting packet details using protocol filters
- Understanding how network communication occurs at packet level

4.Key Features and Benefits Demonstrated

4.1 Deep Packet Inspection

Wireshark enabled detailed inspection of captured packets including source IPs, destination IPs, protocols, ports, timestamps, and packet payloads information.

Observation:

- Packet-level communication between Kali Linux and Windows was successfully analyzed

4.2 ICMP Traffic Analysis

ICMP packets were generated using ping request between Kali Linux and Windows systems.

Activity Performed

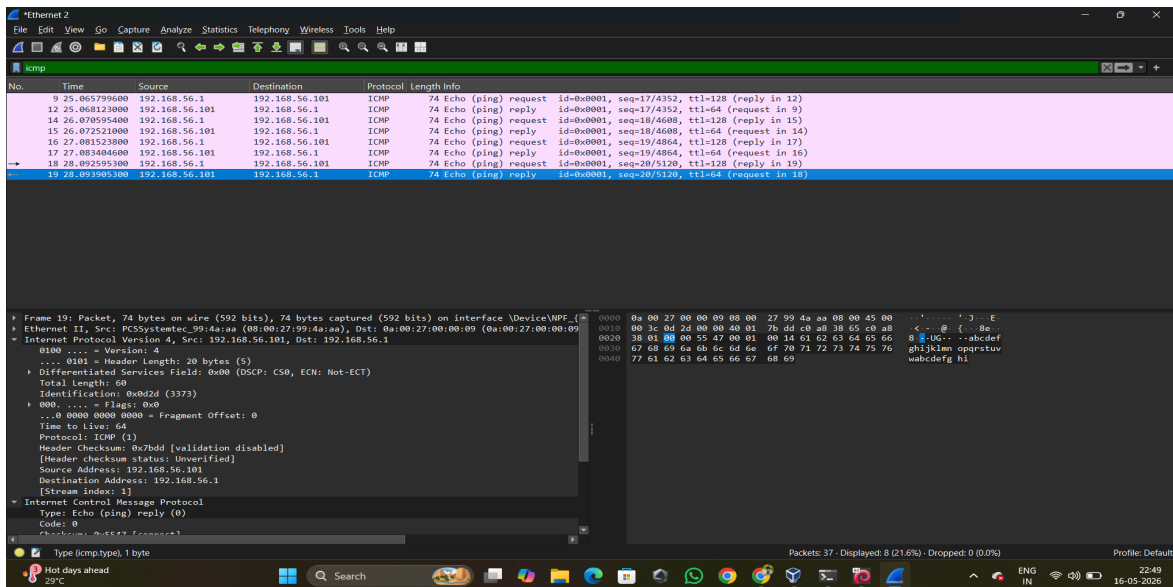
Ping 192.168.56.1

Observation

- Echo Request and Echo Reply packets were captured successfully
- Connectivity between systems was verified

Wireshark Filter

icmp



- ICMP request and reply packets

4.3 TCP Communication Analysis

TCP communication between systems was analyzed to observe session establishment behavior.

Observation

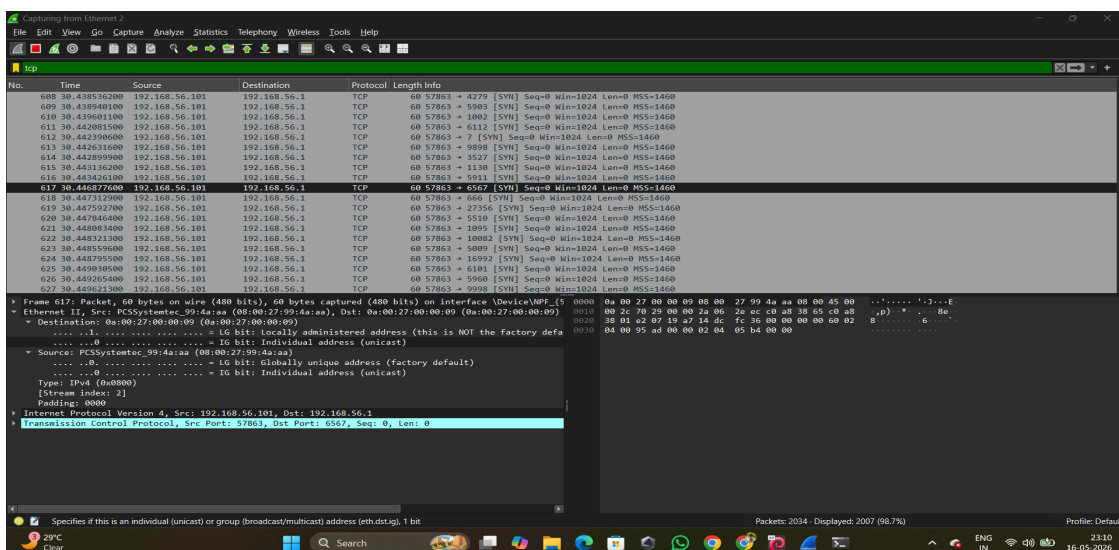
The following TCP handshake process was identified:

- SYN
- SYN-ACK
- ACK

This confirmed successful and reliable communication between Kali Linux and Windows systems.

Wireshark Filter

Tcp



- TCP handshake packets

4.4 DNS Traffic Analysis

DNS traffic was generated to analyze domain name resolution.

Observation

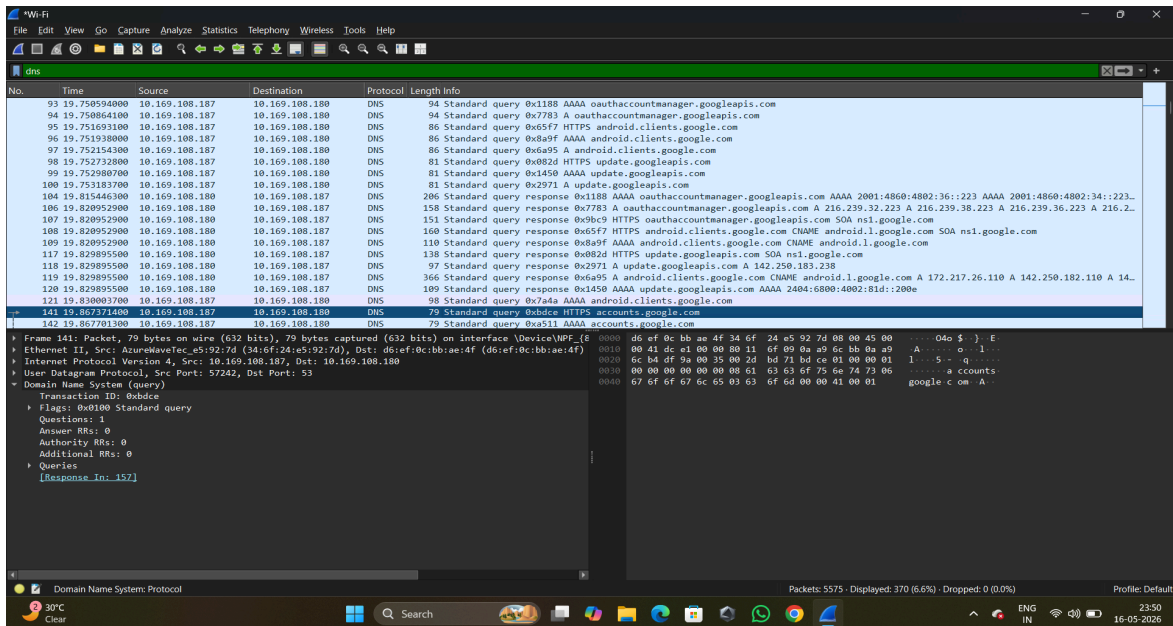
Wireshark captured:

- DNS query packets
- DNS response packets
- Domain-to-IP resolution process

The analysis demonstrated how systems communicate with DNS servers to resolve domains into IP addresses.

Wireshark Filter

Dns



- DNS request and response packets

4.5 HTTP Traffic Analysis

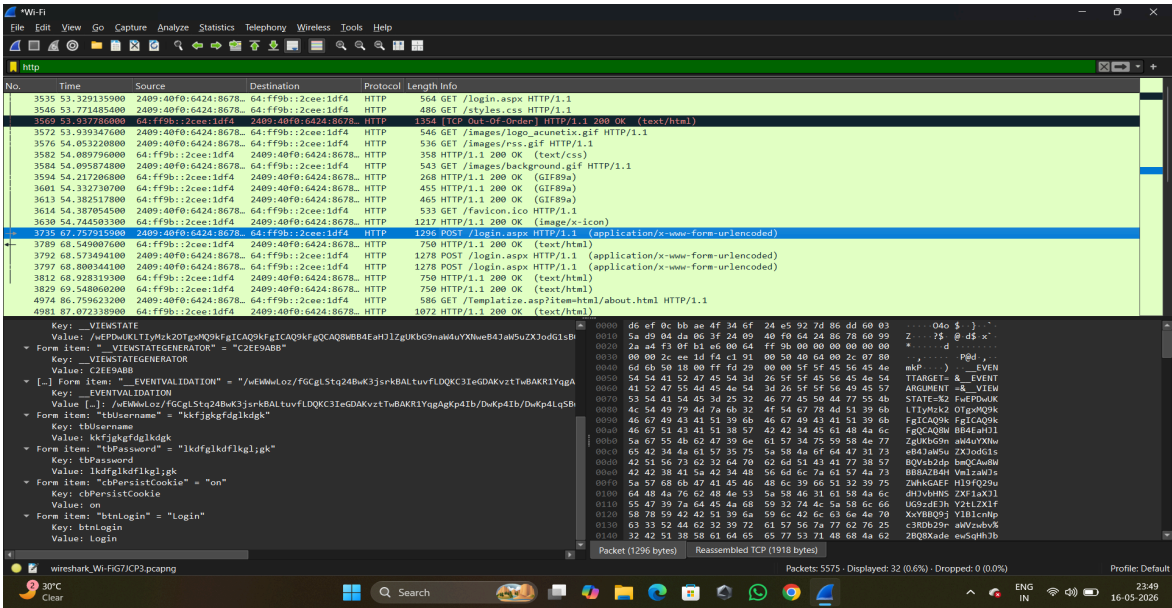
HTTP communicationn was analyzed to inspect plaintext web traffic.

Observation

- HTTP GET requests and responses were captured
- Unencrypted communication was visible in packet contents
- Demonstrated risks associated with insecure protocols

Wireshark Filter

http



- HTTP request and response packets

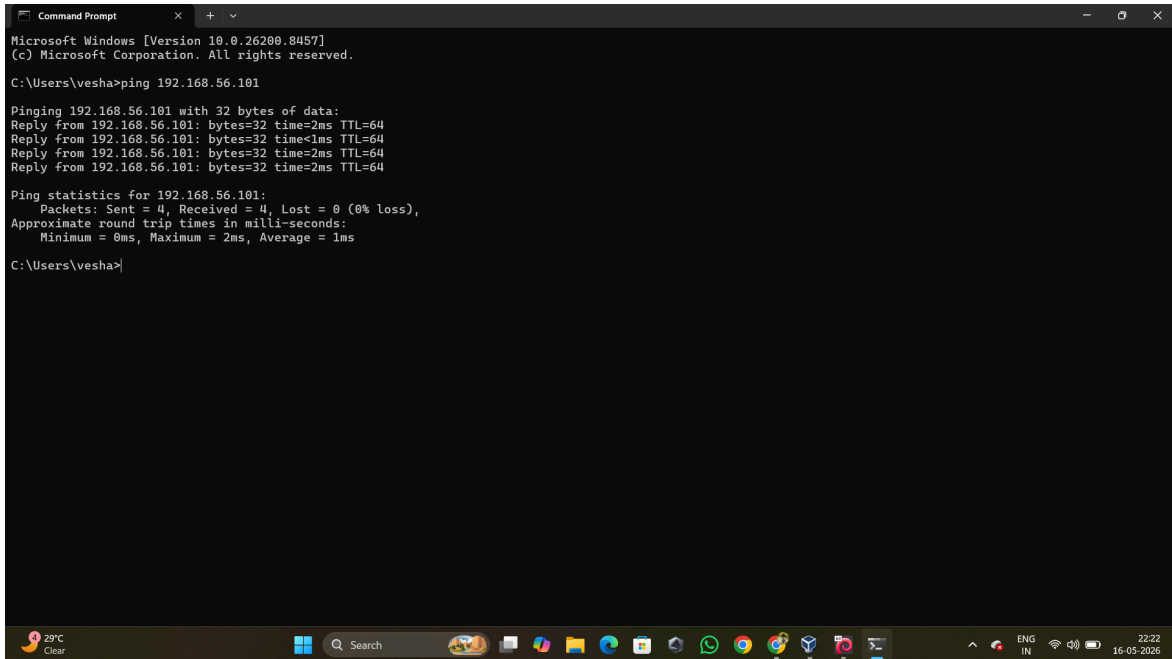
5. Communication Between Kali Linux and Windows

Communication between Kali Linux and Windows systems was continuously monitored during the project.

Observation

- Bidirectional communication was successfully established
- Packet exchange between both systems was verified
- ICMP, TCP, DNS, and HTTP traffic confirmed internal network activity

This demonstrated real-time visibility into internal communication within the virtual lab environment.

A screenshot of a Windows Command Prompt window. The title bar reads "Command Prompt". The window content shows the following text:

```
Microsoft Windows [Version 10.0.26200.8457]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\vesha>ping 192.168.56.101  
  
Pinging 192.168.56.101 with 32 bytes of data:  
Reply from 192.168.56.101: bytes=32 time=2ms TTL=64  
Reply from 192.168.56.101: bytes=32 time=1ms TTL=64  
Reply from 192.168.56.101: bytes=32 time=2ms TTL=64  
Reply from 192.168.56.101: bytes=32 time=2ms TTL=64  
  
Ping statistics for 192.168.56.101:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 0ms, Maximum = 2ms, Average = 1ms  
  
C:\Users\vesha>
```

The taskbar at the bottom shows the Windows logo, a search bar, and various application icons. The system tray on the right shows the date and time as 22:22 on 16-05-2026.

- Windows responding to Linux IPs and giving Reply

7. Outcome of the Project

Network Visibility

Wireshark successfully provided real-time visibility into communication between Kali Linux and Windows systems through packet capture and protocol analysis.

Security Insight

The project demonstrated how unencrypted HTTP communication can be inspected directly using packet analysis tools.

Troubleshooting Efficiency

Packet-level inspection helped verify connectivity, analyze communication flow, and troubleshoot network behavior between systems.

Evidence Collection

Captured packets were stored and analyzed as evidence for understanding communication patterns and protocol behavior.

Operational Benefits

The project improved practical understanding of network traffic analysis, protocol monitoring, and packet inspection techniques relevant to SOC operations and cybersecurity investigations.

8.Conslusion

This project successfully demonstrated the use of Wireshark for network traffic monitoring and protocol analysis in a controlled virtual environment. ICMP, TCP, DNS, and HTTP traffic between Kali Linux and Windows systems were captured and analyzed to understand communication behavior and network visibility.

The project provided practical exposure to packet-level traffic analysis, protocol inspection, and network monitoring techniques commonly used in cybersecurity and SOC environments.

9.Student Details

Field	Details
Student Name	Gurram Harish
Project Title	Network Traffic Analysis Using Wireshark
Tool Used	Wireshark
Virtualization Platform	VirtualBox
Operating Systems Used	Windows and Kali Linux